



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.AT.8

Solve and Graph Inequalities

Lesson 8-6 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Solve

Graph

Solution

Substitute

Solution set

Inverse operation



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

To find all the values that make an inequality true is to

it.

2

To show the solutions on a number line is to them.

3

A value that makes an inequality true is a .

4

To check a solution, I the value back into the inequality.

5

All the values that make an inequality true form the

.

6 An operation that undoes another to isolate the variable is an

Write About the Math

How much more does the detective need? How did you solve and graph this inequality?

¿Cuánto más necesita la detective? ¿Cómo resolviste y graficaste esta desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Solve**
Resolver

☐ **Graph**
Graficar

☐ **Solution**
Solución

☐ **Substitute**
Sustituir

☐ **Solution set**
Conjunto solución

Model: *Solving an inequality gives a range of solutions, not one number.* — Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective needs at least \$__ more because $48 + s \geq 75$ means $s \geq 75 - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$.**
- **My answer is __ because __.**

Mi respuesta es __ porque __.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is __.**
Mi afirmación es __.
- **My evidence is __, so __.**
Mi evidencia es __, entonces __.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Solve
2. **Notes 2:** Graph
3. **Notes 3:** Solution
4. **Notes 4:** Substitute
5. **Notes 5:** Solution set
6. **Notes 6:** Inverse operation
7. **Watch for this mistake:** *A common mistake in Solve and Graph Inequalities is shading the number line in the wrong direction after solving. For example, a student solves $x + 4 > 10$ correctly to get $x > 6$, but then shades LEFT toward smaller numbers instead of RIGHT toward greater numbers. The shading direction always matches the solved symbol, not the original problem: ' $>$ ' and ' $\geq$ ' shade right (greater numbers), ' $<$ ' and ' $\leq$ ' shade left (smaller numbers) — check this AFTER you solve, not before. To catch this mistake, substitute one shaded value and one unshaded value back into the original inequality: the shaded value should make it true, and the unshaded value should make it false.*
8. **Try It 1:** A. Yes, because $5 \leq 5$ is true — Substitute $x = 5$ into $x \leq 5$: $5 \leq 5$ is true because $\leq$ means 'less than OR equal to.' The boundary value 5 IS a solution, so the gate opens.
9. **Try It 2:** A. All numbers greater than 7, like 8, 9, 10, ... (but NOT 7 itself) — The solution set is every value that makes $x > 7$ true. That means all numbers greater than 7, such as 8, 9, 10, and beyond. Because $>$ does not include the boundary, 7 itself is NOT in the solution set.